

System programming question

1. Among the following commands which is used to display the directory attributes rather than its contents?
 - a) ls-l-d
 - b) ls-l
 - c) ls-x
 - d) ls-F
2. Which of the following sits between the user and the Unix Systems?
 - a) CPU
 - b) hardware
 - c) Logic
 - d) Shell
3. Which command is used to display the characteristics of a process?
 - a) ps
 - b) au
 - c) du
 - d) all
4. Which of the following statements are true?
 - a) possible to directly link user-space applications with kernel space
 - b) user-space applications must be allowed to directly execute kernel code
 - c) The C library is at the heart of Unix applications
 - d) All
5. Which UNIX command is used for changing the current file directory?
 - a) rm
 - b) pwd
 - c) cp
 - d) cd
6. Which of the following system calls are used for creating and removing directories?
 - a) rmdir and chown
 - b) mkdir and rmdir
 - c) chdir and chown
 - d) mkdir and chdir
7. Among the following which one is true about hard links?
 - a) hard links allow multiple links map different names to the same inode
 - b) hard links allow for complex filesystem structures with multiple pathnames pointing to the same data
 - c) Hard links cannot span filesystems
 - d) all
8. Which of the following is UNIX character devices?
 - a) hard disks
 - b) CD-ROM drives
 - c) Keyboard
 - d) Flash disk

9. Which of the following is the core of the Linux operating system?
- a) Vi Editor
 - b) Kernel
 - c) Command Line
 - d) Terminal
10. Which is the correct command syntax to locate a file containing the words "hello" and "this"?
- a) locate -i *hello*this
 - b) locate -i **hello**this
 - c) locate -i *hello*this*
 - d) locate -i **hello**this**
11. Which of the following Linux command is used to know which directory you are in?
- a) know
 - b) pwd
 - c) help
 - d) cwd
12. As a system programmer, you want to create a new directory named XYZ, which command will be used for this purpose?
- a) cd XYZ
 - b) mkdir XYZ
 - c) chdir XYZ
 - d) md XYZ
13. How do you show all processes being run by a particular user?
- a) cat /etc/activeprocess | grep USERNAME
 - b) cat /home/USERNAME/open.process
 - c) top -U [USERNAME]
 - d) netstat -u USERNAME
14. When the exit status of the process is undefined?
- a) the main function is not declared to return an integer
 - b) main does a return without a return value
 - c) any of these functions is called without an exit status
 - d) all
15. Which of the following is the advantages of shared libraries in Linux programming?
- a) reduce the size of executable file
 - b) library functions can be replaced with new versions without having to relink edit every program
 - c) a & b
 - d) None
16. Which of the following is not the advantages of malloc function?
- a) easier to use in threaded programs
 - b) provide a simple interface that allows memory to be allocated in small units
 - c) allow us to arbitrarily deallocate blocks of memory
 - d) the allocated memory is initialized

17. _____ a method that kernel uses to minimize the frequency of disk access by maintaining a pool of internal data buffer to increase the response time and throughput?
- a) Pooling
 - b) Buffer cache
 - c) Swapping
 - d) Spooling
18. Which of the following command is used to take the backup in UNIX?
- a) backup
 - b) zip
 - c) gzip
 - d) cpio
19. Which of the following commands will display all the files in your current directory and its subdirectories including the hidden files?
- a) ls -a
 - b) ls -R
 - c) ls -aR
 - d) ls -l
20. Which of the following command displays the current date in the format dd/mm/yyyy?
- a) date +"%d/%m/20%y"
 - b) date +%d/%m/%Y
 - c) date +"%d/%m/%Y"
 - d) date +%d/%m/20%y
21. Which of the following function is used for allocate memory dynamically from the stack?
- a) malloc()
 - b) alloca()
 - c) free()
 - d) none
22. Which of the following is true about the alloca () function?
- a) it allocate block of memory is slower than malloc function
 - b) it allocates is automatically freed when the stack frame is removed
 - c) it needs to maintain a list of free blocks
 - d) none
23. Which of the following is not true about threads in Linux programming?
- a) sharing information between threads is easy and fast
 - b) thread creation is faster than process creation
 - c) a thread uses parallelism which provides a way to improve application performance
 - d) threads are independent of one another like processes
24. What happens when a signal is close to being delivered?
- a) The signal is ignored
 - b) The process is terminated

- c) A core dump file is generated, and the process terminates
 - d) The process's execution is resumed or suspended
 - e) All
25. How to share one thread with other threads within the same process?
- a) thread ID
 - b) code section and data section
 - c) register set and a stack
 - d) program counter
26. Which of the following multithreaded model, the entire process will block if a thread makes a blocking system call?
- a) one to many model
 - b) many to many model
 - c) many to one model
 - d) one to one model
27. Which system call can be used by a parent process to determine the termination of child process?
- a) exit
 - b) wait
 - c) get
 - d) fork
28. Among the following which is used for address of the next instruction to be executed by the current process?
- a) Program counter
 - b) Process stack
 - c) CPU registers
 - d) Pipe
29. Which of the following works by dividing the processor's time?
- a) single task operating system
 - b) kernel
 - c) multitask operating system
 - d) applications
30. Which of the following decides which task can have the next time slot?
- a) single task operating system
 - b) application manager
 - c) multitasking operating systems
 - d) kernel
31. Which of the following provides a time period for the context switch in Linux?
- a) timer
 - b) counter
 - c) time machine
 - d) time slice
32. Where are placed the list of processes that are prepared to be executed and waiting?
- a) Job queue

- b) Ready queue
 - c) Execution queue
 - d) Process queue
33. Which of the following Linux resource limit that controls the maximum number of file locks that a process may hold?
- a) RLIMIT_FSIZE
 - b) RLIMIT_LOCKS
 - c) RLIMIT_DATA
 - d) RLIMIT_MSGQUEUE
34. Which of the following is false about the Virtual memory?
- a) Enables users to run programs that are larger than actual physical memory.
 - b) VM makes the task of programming much easier.
 - c) It allows processes to share files easily and to implement shared memory.
 - d) it uses inefficient mechanism for process creation
35. How do you view the documentation for the command 'kill' in the Linux terminal?
- a) kill -help
 - b) man kill
 - c) kill -help
 - d) doc kill
36. Which of the following userspace is Linux typically paired?
- a) Hurd
 - b) BSD
 - c) GNU
 - d) System V
37. If you want to load the main kernel module for USB support. Which command would help you achieve this task?
- a) modprobe usbcore
 - b) modprobe ehci-ocd
 - c) modprobe uhci-hcd
 - d) modprobe
38. Which directory contains configuration files in Linux?
- a) /bin/
 - b) /dev/
 - c) /etc/
 - d) /root/
39. How many times below program will print hello?
- ```
main()
{
 fork();
 fork();
 fork();
 printf("hello");
}
```
- a) 6

- b) 8
  - c) 3
  - d) 9
40. If you wanted to execute a shell command in the background, which character would you put at the end of the command line?
- a) &
  - b) ;
  - c) :
  - d) @
41. Which of the following is not a feature of UNIX?
- a) Multiuser
  - b) easy to use
  - c) multitasking
  - d) portability
42. Which of the following Linux command-line utility that reports or filters out the adjacent duplicate lines in a file?
- a) paste
  - b) uniq
  - c) sort
  - d) cut
43. Which of the following command is used for searching for a pattern in one or more file(s)?
- a) paste
  - b) cd
  - c) grep
  - d) Cp
44. Which of the following cannot be performed by cat command?
- a) appending files
  - b) creating files
  - c) displaying file
  - d) deleting files
45. Which of the following commands is known as stream editor?
- a) grep
  - b) tr
  - c) sed
  - d) td
46. What is the function of cp command in UNIX?
- a) delete a given file
  - b) change the directory
  - c) list all the available files in the current directory
  - d) cp is a command used for copying files and directories
47. Which of the following is the key feathers of Korn shell?
- a) perform command line editing

- b) maintains command history so that the user can check the last command executed if required
  - c) additional flow control structures
  - d) all
48. Which of the following is commonly used networking commands in UNIX?
- a) sh
  - b) tcsh
  - c) su
  - d) ksh
49. Which of the following UNIX commands that are used for displaying the user information?
- a) nslookup
  - b) tcsh
  - c) Last
  - d) Is -F
50. Which of the following is the features of Stateless Linux server?
- a) Stores prototype of every machine
  - b) Store snapshots
  - c) Store home directories
  - d) All
51. In signal handling, process that receives cannot take the following action.
- a) Ignore the signal such as SIGSTOP and SIGKILL
  - b) Perform system specified default for signal
  - c) Catch the signal, when process catches signal except SIGSTOP and SIGKILL
  - d) None.
52. If the start-up routine were coded in C the call to main could look like
- a) `main(exit(argc, argv));`
  - b) `exit(argc, argv);`
  - c) `exit(main(argc, argv));`
  - d) None
53. What is the major disadvantage(s) of shared library?
- a) Maintaining a single copy of the library routine somewhere in memory that all processes references.
  - b) Reduces the size of each executable file
  - c) Remove the common library routines from the executable file
  - d) none
54. `#include<unistd>` this header file referred by
- a) `void exit (int status);`
  - b) `void _Exit(int status);`
  - c) `void _exit(int status`
  - d) None
55. which could be the proper syntax of initialization function in memory allocation
- a) `void *realloc(size_t, size);`

- b) `void *malloc(void *ptr, size_t newsiz);`
  - c) `void *calloc(size_t nobj, size_t size);`
  - d) All
56. Asynchronous signal/(s)
- a) divided by zero
  - b) keyboard shortcut
  - c) illegal memory access
  - d) All
57. Which one is the valid function that refers to `#include<unistd.h>`
- a) `int rmdir(const char * pathname);`
  - b) `int mkdir(const char * pathname, mode_t mode);`
  - c) `int mkdirat(int fd, const char *pathname, mode_t mode);`
  - d) All
58. From the given options which is unique one from the other
- a) return the last thread from its start routine
  - b) calling abort
  - c) receipt of a signal
  - d) response of the last thread to a cancellation request
59. Which of the following system software resides in main memory always?
- a) Text editor
  - b) Assembler
  - c) Linker
  - d) Loader
60. Daisy chain is a device for?
- a) Interconnecting a number of devices to number of controllers
  - b) Connecting a number of devices to a controller
  - c) Connecting a number of controller to devices
  - d) All of above
61. What is bootstrapping?
- a) A language interpreting other language program
  - b) A language compiling other language program
  - c) A language compile itself
  - d) All of above
62. Shell is the exclusive feature of
- a) UNIX
  - b) DOS
  - c) System software
  - d) Application software
63. A program in execution is called
- A. Process    B. Instruction    C. Procedure    D. Function
64. A UNIX device driver is

- A. Structured into two halves called top half and bottom half
- B. Three equal partitions
- C. Unstructured
- D. None of the above

65. Memory:-

- A. is a device that performs a sequence of operations specified by instructions in memory
- B. is the device where information is stored
- C. is a sequence of instructions
- D. is a computational unit to perform specific functions

66. What will be output of the following code?

```
#include
int main(){
printf(“%d\t”,sizeof(6.5));
printf(“%d\t”,sizeof(90000));
printf(“%d”,sizeof(‘A’));
return 0;
}
```

- A. 8 4 1
- B. 8 4 2
- C. 8 4 4
- D. 8 4 3

67. What will be output of the following c code? ( according to GCC compiler)

```
#include
int main(){
signed x;
unsigned y;
x = 10 +- 10u + 10u +- 10;
y = x;
if(x==y)
printf(“%d %d”,x,y);
else if(x!=y)
```

```
printf(“%u %u”,x,y);
return 0;
}
```

- a. 0 0      b. 65536 -10      c. 0 65536      d. Compilation error

68. What will be output of the following c code?

```
#include
int main(){
const int *p;
int a=10;
p=&a;
printf(“%d”,*p);
return 0;
}
```

- A. 0      B. 10      C. Garbage Value      D. Any Memory address

69. Which of the following is integral data type?

- A. void      B. char      C. float      D. double

70. Expansion time variables are used

- A. Before expansion of micro calls  
B. Only during expansion of macro calls  
C. After expansion of micro calls  
D. Any one of the above

71. A model statement contains call for another macro is called as

- A. referential macro call  
B. nested macro call  
C. inbuilt macro call  
D. inherited macro call

72. The flow control during macro expansion is

- A. combination      B. chronological      C. indexable      D. sequential

73. A \_\_\_\_\_ statement declare the name of macro.
- a) macro prototype
  - b) macro definition
  - c) macro identification
  - d) None of the above
74. A macro definition consists of
- A. A macro prototype statement
  - B. One or more model statements
  - C. Macro pre-processor statements
  - D. All of the above
75. The translator which perform macro expansion is called a
- A. Macro processor
  - B. Macro pre-processor
  - C. Micro pre-processor
  - D. assembler
76. Which command is used with vi Editor to delete a single character
- A. x
  - B. y
  - C. a
  - D. z
77. What command is used to count the total number of lines, words, and characters contained in a file?
- A. countw
  - B. wcount
  - C. wc
  - D. count p
78. How can you add “Zerihun”, a new user, to your system?
- A. Using useradd
  - B. Using adduser
  - C. Using linuxconf
  - D. All
79. What file specifies the order in which to use specified name services?
- A. /etc/services
  - B. /etc/nsorder
  - C. /etc/nsswitch.conf
  - D. /etc/hosts
80. How many primary partitions can exist on one drive?
- A. 16
  - B. 4
  - C. 2
  - D. 1
81. In which directory can you store system user default files used for creating user directories?
- A. /usr/tmp
  - B. /etc/default
  - C. /etc/skel
  - D. etc/users
82. Which is the correct matching?
- A. TIME\_OOP : a leap second just occurred

- B. TIME\_WAIT : a leap second is in progress
- C. TIME\_BAD : the clock is partially synchronized
- D. TIME\_OK : the clock is synchronized

83. How to locate and get the process time or in which header file?

- A. #include<sys/times.h>                      C. #include<times.h>
- B. #include<utils/times.h>                      D. #include<clocks.h>

84. Which is not true about the times () system call which retrieves the process time of the running process and its children, in clock ticks?

Struct tms {

}; clock\_t times(struct tms\*buf);

- A. Clock\_t tms\_ftime; // user time consumed
- B. Clock\_t tms\_stime; // system time consumed
- C. Clock\_t tms\_cutime; //system time consumed by children
- D. Clock\_t tms\_usertime; //user time consumed by children

85. The simplest interface to the signal features of the UNIX System is the signal function.so, which could be the correct syntax ?

- A. void (int) (\*signal(int signo, void (\*func)(int)))(int);
- B. void (void (\*signal(int signo, void (\*func)(int)))(int))(int);
- C. void (\*signal(int signo, void (\*func)(int)))(int);
- D. None

86. Which is not true about closing the opened file?

#include <fcntl.h>

int close(int fd);

- A. fd: file descriptor    B. 0 :on success    C. -1 : on error    D. NULL : on error

87. How to create threads?

#include<pthread.h>

- A. `int pthread_create(pthread_t *restrict tidp, const pthread_attr_t *restrict attr, void *(*start_rtn) (void*), void *restrict arg);`
- B. `pthread_t pthread_self(void);`
- C. `int pthread_equal(pthread_t tid1, pthread_t tid2);`
- D. `pthread_t create(thread tid1);`

88. Which is the correct signature to exit thread?

`#include<pthread.h>`

- A. `int pthread_exit();`
- B. `void pthread_exit(void*rval_ptr);`
- C. `void pthread_exit(void);`
- D. `int pthread_exit(0);`

89. From the following given options one is not describe about signals?

- A. software interrupts
- B. most nontrivial application programs need to deal with signal
- C. signals provide a way of handling asynchronous events
- D. signals have the capacity to protect the system

90. true about signals:

- a) SIGFPE : bus error
- b) SIGCONT : restart
- c) SIGILL : interrupt the process
- d) SIGINT : division by zero

91. From the following which could be fully describe about Application Programming Interface (API)?

- a) The interface by which one piece of software communicates with another at the source level
- b) It provides abstraction by providing a standard set of interfaces at low level piece
- c) Binary interface between two or more pieces of software on a particular architecture
- d) They are concerned with issues such as calling conventions, byte ordering, register use, system call invocation.

92. Which is the correct scenario about POSIX?

- a) Richard Stallman suggested the standard be named POSIX
- b) Tim-Berners Lee was published the first POSIX Standard
- c) Daphne Koller is the first linux standard inventor

- d) Geoffrey Hinton who was the first unix standard inventor
93. Which are not correct primitive system data types ?
- A. mode\_t      B. cclock\_t      C. size\_t      D. ssize\_t
94. the primitive system data type which is used to define the file sizes and balances
- A. gid\_t      B. ino\_t      C. off\_t      D. pid\_t
95. in which of the following header file the system primitive system data types found?
- A. <sys/types.h>      B. <types.h>      C. <util.h>      D. <ctype.h>
96. Which is the correct signature of symbolic link?
- a) int symlink(const char \*pathname, const char \*slink);
  - b) void symlink(const char \*pathname, const char \*slink);
  - c) char symlink(const char \*pathname, const char \*slink);
  - d) string symlink(const char \*pathname, const char \*slink);
97. What is the difference between hard and symbolic link ?
- a) A soft link is a file all its own, and the file references or points to the exact spot on a hard drive where the Inode stores the data.
  - b) A hard link isn't a separate file, it points to the name of the original file, rather than to a spot on the hard drive
  - c) soft link is similar to file shortcut where as hard link is copy of original file
  - d) A and B
98. Which are not Valid optional constants of open() and openat() system:
- a) O\_APPEND
  - b) O\_CREAT
  - c) O\_NOFOLLOW
  - d) O\_ASYNCNO
99. How to use lseek function?
- A. <unistd.h>    off\_t    lseek(int fd,    off\_t    offset,    int    whence);
- B. <stdio.h>    offset\_t    lseek(int fd,    off\_t    offset,    int    whence);
- C. <apue.h>    offset\_t    lseek(int fd,    off\_t    offset,    int    whence);
- D. <types.h>    int    lseek(int fd,    int    offset,    int    whence);
100. How many data structure uses the kernel to represent an open file?
- A. 1                      B. 2                      C. 3                      D. 4